

Round to a place value



1 Round the following to the nearest whole number:

- | | | | |
|--------|--------|---------|----------|
| a 2.19 | b 5.97 | c 24.09 | d 33.931 |
| e 0.5 | f 0.05 | g 0.799 | h 89.5 |

2 Round the following to the nearest tenth:

- | | | | |
|---------|-----------|----------|---------|
| a 0.14 | b 7.35 | c 3.56 | d 8.99 |
| e 27.88 | f 289.427 | g 0.7003 | h 9.992 |

3 Round the following to the nearest hundredth:

- | | | | |
|---------|-----------|----------|-----------|
| a 2.897 | b 328.864 | c 14.285 | d 67.001 |
| e 0.007 | f 12.999 | g 99.999 | h 131.099 |

Round to a number of decimal places

4 Round the following to one decimal place:

- | | | | |
|--------|---------|---------|----------|
| a 0.41 | b 8.35 | c 79.38 | d 0.3009 |
| e 3.55 | f 19.95 | g 13.88 | h 1.0099 |

5 Round the following to two decimal places:

- | | | | |
|----------|-----------|-----------|----------|
| a 2.697 | b 393.586 | c 0.435 | d 7.3582 |
| e 79.997 | f 2.0029 | g 149.955 | h 0.099 |

6 Round the following to three decimal places:

- | | | | |
|-----------|------------|------------|------------|
| a 11.8483 | b 2.159 76 | c 4.504 13 | d 1.599 64 |
| e 90.0335 | f 0.5987 | g 41.5005 | h 9.9996 |

7 Round the following to four decimal places:

- | | | | |
|------------|------------|-------------|-------------|
| a 5.659 64 | b 5.600 96 | c 12.006 65 | d 39.999 95 |
|------------|------------|-------------|-------------|



Recurring decimals

- 8 The fraction $\frac{1}{9}$ can also be written as $1 \div 9$.

The quotient is $0.111 \dots$ which we read as "zero point one recurring".

$0.111 \dots$ can be written as $0.\dot{1}$.

Complete the table for the following recurring decimals:

Fraction	Recurring decimal	Rounded to two decimal places
$\frac{1}{9}$	$0.\dot{1}$	0.11
$\frac{2}{9}$	$0.\dot{2}$	
$\frac{3}{9}$		
$\frac{4}{9}$	$0.\dot{4}$	
$\frac{5}{9}$	$0.\dot{5}$	
$\frac{6}{9}$	$0.\dot{6}$	
$\frac{7}{9}$	$0.\dot{7}$	
$\frac{8}{9}$		0.89
$\frac{9}{9}$	1	1.00

- 9 The fraction $\frac{1}{7}$ can also be written as $1 \div 7$.

The quotient is $0.142857142857 \dots$ when written as a decimal, which can be written as $0.\dot{1}4285\dot{7}$.

Complete the table for the following recurring decimals:

Fraction	Recurring decimal	Rounded to four decimal places
$\frac{1}{7}$	$0.\dot{1}4285\dot{7}$	0.1429
$\frac{2}{7}$	$0.\dot{2}8571\dot{4}$	
$\frac{3}{7}$		
$\frac{4}{7}$	$0.\dot{5}7142\dot{8}$	
$\frac{5}{7}$	$0.\dot{7}1428\dot{5}$	
$\frac{6}{7}$	$0.\dot{8}5714\dot{2}$	
$\frac{7}{7}$	1	1.0000

10 Round the following recurring decimals to  decimal place:

a $0.\dot{2}4$

b $0.\dot{6}3\dot{1}$

c $0.\dot{4}$

d $2.\dot{6}$

e $0.\dot{1}\dot{9}$

f $0.\dot{8}0\dot{8}$

g $2.\dot{5}$

h $4.2\dot{6}$

11 Round the following recurring decimals to two decimal places:

a $0.\dot{1}3\dot{8}$

b $0.\dot{8}9$

c $0.\dot{7}$

d $9.\dot{9}$

e $0.\dot{7}\dot{2}$

f $0.\dot{5}8\dot{6}$

g $0.\dot{4}$

h $3.\dot{5}$